Summary sheet

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Internship topic

Data-driven inference of unsteady inlet conditions for the numerical simulation of the BARC test case.

Objectives

Numerical simulations will be carried out in order to obtain a comparison between experimental and numerical results. First of all, multiple simulations will be carried out with different turbulence models, using the low turbulent intensity configuration. First, the aim is select the most adapted turbulence model which fits the CSTB experimental studies.

On the BARC test case, the inlet boundary is very sensitive. It is thus a study of interest to introduce DA to predict and optimize the turbulence production's parameters, in the case of the high turbulence intensity configuration. The aim to use DA to numerically optimize some inlet boundary parameters and get some flow characteristics which were also obtained during the experiments led by CSTB, such as the recirculation region's size.

Main customer

Not relevant

Used tools

- Linux OS
- OpenFOAM (v8, v9, v2206)
- High Performance Computing
- Python

Conducted studies

On the BARC test case, the following studies have been carried out:

- Acquaintance with existing bibliography.
- Comparative study between several Spalart-Allmaras turbulence models.
- Using the SA-DDES turbulence model, study of the mesh influence.
- Frequency analysis on the shear layer over the body's upper surface.
- Establishment of a data assimilation routine using the Ensemble Kalman Filter.
- Development of a post-processing routine.
- Several prediction and analysis phases have been launched.

Results

The results are:

- The best recirculation region was obtained using the SA-DDES turbulence model, on a baseline mesh composed of 3 million mesh elements.
- Two refined meshes were implemented, with 4 and 30 million mesh elements. However, both refined meshes yielded to the same recirculation region. The 4 million element grid is then preferred.
- The mesh has a strong influence on the results, especially in regions close to the body.
- Data assimilation yielded to a recirculation region whose length has been optimized, and went from a size of 59 to nearly 67% of the body's length. This was obtained thanks to four prediction and analysis steps.

Explanation of possible differences

Possible differences might come from:

- Numerical errors.
- A mesh which might be still too coarse.
- Simulations that did not perfectly converged.
- A turbulence model which might not be the most adapted for our case study.
- A bad choice in the data assimilation routine, especially during the observation sets on which the predictions will be projected on.
- Once again, ensembles might not have converged and yielded to approximated results.

Encountered difficulties

The difficulties were the following:

- Getting started with the Linux OS environment.
- Developing post-processing routines using Python.
- Perfectly understanding how the Ensemble Kalman Filter works.

This difficulties have been overtaken thanks to loads of research, asking help, and perseverance.

Work to be continued

The following tasks need to be continued:

- Thanks to the optimized parameters obtained with the data assimilation work, a new simulation will be launched from scratch, the goal being to obtain a very resolved numerical solution close to the experimental results obtained by CSTB.
- An article in a international review is planned to be written.